

## Counting little fish for science

**Sponsor Name:** Dennis Cocherell and Florian Mauduit Wildlife, Fish, and Conservation Biology  
Dept. University of California, Davis

**Contact Information:** decocherell@ucdavis.edu and fmauduit@ucdavis.edu

### Project Motivation:

We are looking for a means to count fish passing through our fish friendly transfer pump (<https://www.faivre-aqua.fr/en/produit/pescamotion-30-6-fishing/>). We rear endangered Delta smelt, and other species, that get transferred from rearing tanks to a transport truck for release into experimental ponds or supplementation in the Sacramento river. We currently hand count the fish into a 20' tank (~20,000) or into a wild type pond (~3000) prior to release. Unfortunately, we can not net the fish to count them at this stage as the fish are extremely sensitive to handling. To be able to gauge success of our rearing environments we need to be able to count the fish.

### Objective:

Develop a method and/or device (video, laser??) to enumerate fish passing through a 4" hose, the hose is variable in length from 20 to 40' long.

### Key Design Considerations:

Ability to accurately count fish moving at speed through a 4" hose; either prior to entering, on the suction side of the pump, the discharge side of the pump or the outlet

- Be able to discard data or filter data for bubbles or debris (post-processing??) and within semi-turbid water
- Thousands of fish can pass through within a few minutes, be able to keep pace
- Be able to work in a wet environment
- Be able to mount device inline along the route of the fish passing through the system

### Additional Information:

This website is for the pump; <https://www.faivre-aqua.fr/en/produit/pescamotion-30-6-fishing/>. This youtube video show the pump in operation at one of our facilities;

<https://www.youtube.com/watch?v=i4kL7tYhs40> This is an article a little about the supplementation program for the fish reared at our UCD facility; <https://mavensnotebook.com/2024/01/29/notebook-feature-ramping-up-releases-of-hatchery-delta-smelt-to-the-wild/>

### Figures or Graphics:

We can conduct a short powerpoint if there is interest in this project.

